QP CODE:24900271

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER MGU-UGP (HONOURS)
REGULAR EXAMINATION NOVEMBER 2024
First Semester

Discipline Specific Core Course : **MG1DSCZIM101- INTRODUCTION TO MICROBIAL WORLD**
(2024 Admission onwards)

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall Outcome attainability

Part A

Fill in the blanks. Answer **All** question
Each question carries **1** marks

1. ……. is the father of Microbiology [K] [1]
   Ans: Antonie Van vanleeuwnhoek
2. The strong belief that existed among the scientists at that time in the theory of …… [E] [1]
   Ans: Spontaneous generation
3. The bacterial cell wall is composed of a layer of …… [A] [2]
   Ans: Peptidoglycan
4. …… are proteins found in the outer membrane of certain bacteria and mitochondria [U] [2]
   Ans : Porins
5. Bacteriophage were first discovered by [S] [3]
   Ans: Fredrick Twort
6. The term Bacteriophage literally means [S] [3]
   Ans: Bacteria eater
7. Hyphae are divided into …… by septa [U] [4]
   Ans: Cells
8. Fungi reproduce through……. [A] [4]
   Ans:Sexually or asexually
9. Protozoa play a crucial role in…… [U] [5]
   Ans:Water Purification
10. Protozoa reproduce …….. through binary fission [An][5]
    Ans:Asexually

(10x 1 =10 Marks)

Part –B

Short Questions. Answer any **10** question
Each question carries **2** marks

11.Define Microbiology **[K] [1]**

Microbiology is the study of very small living organisms that can be viewed only with a microscope.

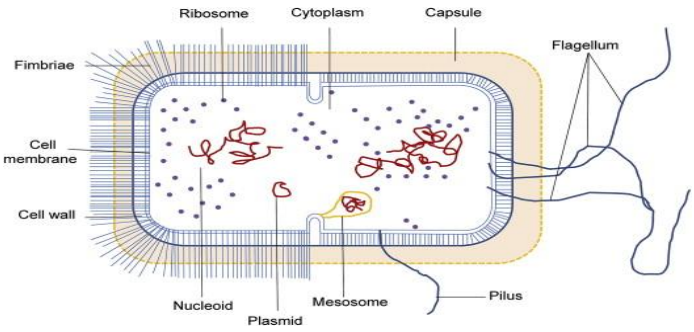
12.Define golden age of microbiology **[K] [1]**

The period of discoveries is known as golden age of microbiology.

13.Discuss Koch postulates **[U] [1]**

- The organism should always be associated with a particular diseases.The organism should be isolated from the pathological specimen and grown in artificial culture medium in the laboratory.
- The pure culture of an organism when inoculated into an animal should produce the same type of diseases.From the experimentally infected animal the same organism should be re-isolated in pure culture**.**

14. Describe the morphology of bacteria with diagram **[U] [2]**



15.Write short note on gram positive cell wall **[A] [2]**

It is composed of peptidoglycan which contains N- acetyl muramic acid and N – acetyl glucosamine molecules.

16.Define the features of any two internal structure of bacteria **[U] [2]**

Cytoplasm :- It is athick watery solution

.**Ribosomes :-** It is the centre of protein synthesis.

Nucleoid :- It contains a single, long circular double stranded DNA

Inclusion bodies :- It is the reserve deposits of excess nutrients

Mesosomes :- It is the main sites of respiratory enzymes**.**

17.Write the history about Bacteriophage **[U] [3]**

Bacteriophage are viruses that infect bacteria. The bacterial hosts can be easily grown and infected with phage in the laboratory.Bacteriophages have been extensively used in the study of bacterial genetics and cellular control mechanisms.

18.Write down the steps involved in the lytic cycle **[A] [3]**

Adsorption, Penetration, Synthesis,Maturation and Release o f phage particles

19.Write short note on fungi **[U] [4]**

- Fungi are eukaryotic organism .Fungi have rigid cell walls containing chitin , mannan and other polysacharides.Presence of true nucleus fungi.They produce sexually and asexually

20.Distinguish the characteristic features of Ascomycota **[U] [4]**

•The hyphae are usually septate.They form both sexual and asexual spores
•The sexual spores are known as ascospores. Example :- Aspergillus, Pencillium

21.Explain the important features of protozoa [U] [5]

Ecological imlportance
Environmental importance
Agricultural Importance

22.Explain the significance of cilia, flagella and Pseudopodia in protozoa [U] [5]

Flagella :- The flagellum is an extremely fine filamentous extension of the cell .
Cilia :-It is a locomotory organ.
Pseudopodia :- A pseudopodia is a temporary projection of part of the cytoplasm

(10 x 2 = 20 Marks)

Part – C

Short Essay Type Question. Answer **5** Questions
Each Question carries **4** Marks

23.Describe the history of Microbiology. [U] 1]
(Mention about any 4 scientists and their contributions-1 Mark each)

- Period of Controversies
- Period of discoveries
- Period of modern Microbiology

24. Explain the experiment which disprove the theory of spontaneous generation. How?

(Francesco Redi-1 Mark,Experiment-2 Marks,diagram-1 Marks) [U] 1]

- Redi took two jars and placed a piece of meat in each.He covered the mouth of one jar with apiece of gauze and left the other jar with a piece of guaze and left the other openFlies entered the open jar sat on the meat and laid eggs from which maggots appeared . in the closed jar , flies could note reach the meat and so maggots did not appear.Redi thus proved that maggots did not appear of their own but orginated because of the flies

25. Illustrate the ultrastructure of bacteria and their functions [A] [2]

(External & Internal structures-2 Marks, Diagrammatic representation-2 Marks)

Fimbriae and pili :-Short, thin ,straight hair like appendages on the cell wall of bacteria.**Flagella** :- It is long, semi rigid, hollow, tubular structure attached to cell membrane. **Capsule** :- It is a sticky bacterial secretion that forms an envelope around the cell wall.**Cell wall** :- It give the shape and rigidity to the cell. **Cytoplasmic membrane :-** It is a thin layer that lies below the cell wall enclosing the cytoplasm.
Internal structures

Cytoplasm :- It is athick watery solution containing a number of organic and inorganic substances . **Ribosomes** :- It is the centre of protein synthesis.
Nucleoid :- It contains a single, long thread of double stranded DNA .**Inclusion bodies** :- It is the reserve deposits of excess nutrients.**Mesosomes** :- It is the main sites of respiratory enzymes.

26. Discuss the morphological features of Bacterial cell walls [U][2]
(Peptidoglycan-1 Mark,Gram positive-1.5 Marks,Gram negative-1.5 Marks)

- The cellwall gives shape and rigidity to the cell. It protects the internal structure
- It is composed of peptidoglycan which contains N- acetyl muramic acid and N – acetyl glucosamine molecules.The cell wall of gram negative bacteria is chemically more complex than that of gram positive bacteria

27. Explain the detailed note on Bacteriophage [U] [3]
(Definiton-1 Mark,Diagram-1 Mark,Structure-2 Marks)

Bacteriophage are viruses that infect and parasitize bacteria. A typical phage particle is tadpole shaped consisting of a head and a tail.it consist of a tightly packed core of a nucleic acid, DNA or RNA enclosed by a protein coat called capsid.

28. Distinguish the characteristic features of Ascomycota and Basidiomycota (Ascomycota-2 Marks,Basidiomycota-2 Marks) [U] [4]

Ascomycota : The hyphae are usually septate.They form both sexual and asexual spores.The sexual spores are known as ascospores.These are formed by the fusion of two nuclei of either morphologically similar or dissimilar cells.

Basidiomycota: These are also known as club fungi.They usually possess septate hyphae

29. Discuss the ecological importance of Protozoa. [U] [5]
(Mention any 4 importance -1 Mark Each)

- Soil Fertility it help maintain soil fertility by regulating bacterial populations and enhancing the rate at which bacteria decompose dead organic matter
- Bioindicators - Protozoa can be used as indicators of changes in the environment
- Protozoa play critical roles in ecosystem serving as primary producers, decomposers and important links in food chain

(5 × 4 = 20 Marks)

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